

MD. AL AMIN CHY

Prospective Graduate Student

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WORK EXPERIENCE

1) Adjunct Lecturer

Dept. Of Electrical and Electronic Engineering, Sylhet Engineering College.

Aug '25 — Present
Full Time, Bangladesh

- Taken undergraduate courses including **Theory: 1) *Electromagnetic Fields & Waves*, 2) *Electrical Properties of Materials*, and 3) *Electronics II*, lab: 4) *Measurement & Instrumentation Lab* and 5) *Communication I*. [Certificate](#)**
- Deliver lecture sessions, graded assignments, and provided one-on-one academic support to students.

2) Research Assistant

Dept. Of Electrical and Electronic Engineering, Shahjalal University of Science and Technology

May '26 — Present
Part Time, Bangladesh

- Applied FDTD, EMT, and TMM techniques to analyze optical responses, resonance behavior, sensing performance.
- Designed ultra-broadband metamaterial absorbers for solar energy harvesting and thin-film photovoltaic applications.

3) Research Assistant

Faculty of Agriculture Engineering, Sylhet Agriculturer University.

Jan '25 — Jun '25
Internship, Bangladesh

- Simulated using ANSYS HFSS and COMSOL Multiphysics, with hands-on experience in photonics labs and optical materials. [Certificate](#)
- Conducted numerical analysis, RF/microwave simulations, and research manuscript preparation related to light-matter interactions and electromagnetic structures.

PUBLICATIONS

→ **Md. Al Amin Chy**, TH Tasneem, SA Chowdhury, MRC Mahdy. arXive. "Ultra- Broadband Electromagnetic Control Using a Circular Ring Metasurface: Surface Wave Propagation, Beam Steering, and RCS Reduction (50 - 100 Ghz)". [Live](#)

→ **Md. Al Amin Chy**, M.A.H. Sakib Opu, M.J.A. Soeb. ICERIE, 2025. Advance Engineering Research, part of Springer Nature "*Optical Sorting of Distinct Nanorods Chiral Dielectric Hyperbolic Metamaterial in Clustered Configurations*". [Live](#)

→ **Md. Al Amin Chy**; Sabbir Ahmad Chowdhury; Rashedul Albab; Tasnimul Hasan Tasneem; Mizanur Rahman; Arif Ahammad. ICCIT(IEEE),2025, "*Time Dependent Control of Voltage and Current via a Tunable Metasurface*" [Live](#)

→ M.F. Ahmad, Rashedul Albab, **Md. Al Amin Chy**, M.A.H. Sakib Opu, Tahsin Hadi, Tanvir Ahamed. ICCIT(IEEE), 2025, "*Explainable Machine Learning for Multi-Class Power Quality Disturbance Classification Using SHAP and Feature Importance Analysis*". [Live](#)

→ M.F. Ahmad, R. Albab, M.A. Ahmad, **Md. Al Amin Chy**, S.N. Mehdi. ICCIT (IEEE), 2025, "*Quantifying the Relationship between Socioeconomic Status and Parent-Child Attachment on Adolescents through Machine Learning*". [Live](#)

RESEARCH EXPERIENCES

⇒ "Ultra-Broadband Electromagnetic Control Using a Triple Circular Ring Metasurface: Surface Wave Propagation, Beam Steering, RCS Reduction (50-100 Ghz)", Supervised by [Professor Dr. Mahdy Rahman Chowdhury](#) (Photonic Time Crystal, Metasurface) January 2024 - December 2024. [Project](#)

- Phase 1 (6 months):** Studied metasurface and photonic time crystal physics; reproduced published results and compiled research notes.
- Phase 2 (6 months):** First author on a journal manuscript; designed and optimized a multifunctional metasurface in CST (100+ simulations) and prepared figures using MATLAB and BLENDER.
- Research Recognition:** Received a [supervisor-certified recognition letter](#) for successfully completing and submitting the research project for Q1 journal publication consideration under his supervision.

⇒ "Robust Optical Pulling of Plasmonic, Chiral, and Dielectric Nanorods via Hyperbolic Metamaterials",

Supervised by [Assistant Professor Md. Janibul Alam Soeb](#)

(Optical Manipulation, Metamaterial, Plasmonics) January 2025 - June 2025. [Project](#)

- **Conceptualization & Simulation (3.5 months):** Studied relevant papers for idea development and physics understanding, reproduced published results, selected suitable materials, designed optical setups, and performed force calculations using COMSOL Multiphysics.
- **Scientific Paper Writing (2.5 months):** First author on the manuscript; contributed to methodology, simulation analysis, results interpretation, and scientific writing under him.

EDUCATION

B.Sc. (Engineering) in Electrical and Electronics Engineering, Shahjalal University of Science and Technology, Bangladesh. (GPA: 3.88/4.0 with distinction) Jan '20 — Jul '25
Sylhet, Bangladesh

- Advanced Calculus, Differential Equations, Linear Algebra, Complex Variables, Embedded systems.
- Signals Processing, Analog & Digital Electronics, Electromagnetics, Quantum Mechanics, Solid State Device, RF & microwave, Materials, Communication, Renewable Energy, Power electronics.

SKILLS

CST Microwave Studio, COMSOL Multiphysics, ANSYS HFSS, Python, MATLAB, Microwave devices and passive components, RF circuit analysis and design, LaTeX, C++, Microsoft Excel, Powerpoint, Teamwork, FDTD, Lumerical

PROJECTS

- 1) Three-Phase Inverter System using IGBT and Arduino**, Power Electronics Lab Jan '25
 - Designed a DC–AC inverter using IGBT with SPWM switching control, for variable-frequency motor drives
- 2) Line Following Robot**, Technoventure 2.0 Robotics Competition [Link](#) Jul '23
 - Built an autonomous robot using sensors and microcontroller-based control. [Link](#)
- 3) Feasibility Study of a Renewable Energy Project using SAM**, Renewable Energy System Lab Dec '24
 - Conducted technical and financial assessment of solar PV and wind systems using SAM. [Link](#)

AWARDS & ACHIEVEMENTS

- 1) Academic Distinction Award** Jul '25
Shahjalal University of Science and Technology, I am proud to be ranked among the top 3% among others. [Link](#)
- 2) Merit-Based Scholarship** Jun '25
Shahjalal University of Science and Technology, Consistently received throughout all semesters of the program. [Link](#)
- 3) Best Campus Ambassador** Jan '24
Bangladesh Environment Olympiad 2024, Awarded for environmental awareness outreach. [Link](#)

CO-CURRICULAR ACTIVITIES

- 1) Organizing Secretary**, Aowlasher Sonal Welfare Organization (Govt.-registered, Reg. No. 869) Oct '21 — Sep '23
Responsible for coordinating organizational activities. [Link](#)
- 2) Publicity Secretary**, Electrical and Electronic Engineering Association Mar '24 — Jun '25
Responsible for publicity campaigns and communication. [Link](#)
- 3) Champion (2 times)** Dec '22 — Dec '23
Tournaments: 1) Engineer Football League, 2) Estrada: The Glory Indoor Football. Played as Left Forward. [Link](#)